Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/07/2007
PI Name:	Lathan, Corinna E. Ph.D.		
Project Title:	Mental Representation of Spatial Cues (3D-Space)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	clathan@atinc.com	**Fax:**	FY 301-585-9075
PI Organization Type:	INDUSTRY	**Phone:**	301-495-0770
Organization Name:	AnthroTronix, Inc.		
PI Address 1:	8737 Colesville Rd. L-203		
PI Address 2:			
PI Web Page:	www.atinc.com		
City:	Silver Spring	**State:**	MD
Zip Code:	20910	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/02/2005	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Unassigned Co-Investigator to ESA managed flight study that is in flight development phase.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clement, Gilles (CNRS (Centre national de la recherche scientifique))		
Grant/Contract No.:	NNJ05HD92A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The purpose of this study is to investigate the effects of acute and long-term effects of exposure to microgravity on the mental representation of spatial cues by astronauts during and after space flight. Previous space studies carried out by our group have shown that cognitive functions such as mental rotation, symmetry detection, and the internal representation of the vertical axis of objects were found to be impaired during and after space flight. We propose to repeat the preliminary observations on a larger number of human subjects during short-and-long-term space missions, by using simple experiments requiring a laptop computer and a digitizing table. The analysis of the spatial and dynamic components of writing and drawing tests of 2-D and 3-D objects performed throughout a space mission should allow the researchers to confirm that the internal spatial representation of the vertical dimension, which is determined on Earth by the gravitational reference system, is altered during space flight. If crew time and equipment availability permit, additional psychophysics measurements will be made using a portable virtual reality system displaying 3-D objects with one dimension altered. Tests of the perception of absolute distances will also be performed using the same equipment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study could have important consequences for human performance during space flight. For example, if an astronaut imagines a space station shorter than it is in reality, the expected duration of a trip within this station will be shorter than what it actually takes, thus leading to delays and stress. There may also be consequences for a space habitat design if squared volumes do not look square to astronauts. Virtual reality training may be a way to train the astronauts to compensate for such altered spatial representation.
Task Progress:	During the current reporting period 1791 hours were billed to this effort. Software to support the assessment of mental representation of spatial cues during spaceflight was further developed and tested in ground-based and parabolic flight experiments. The final version of the software is currently being transitioned to ESA. The associated hardware for use in spaceflight was selected, tested in ground-based and parabolic flight experiments, and is currently being qualified for spaceflight by CNES. Additionally, experimental design and administrative tasks were completed during this period of performance. Specific tasks performed are detailed in the sections to follow. Hardware Evaluation and Configuration: Work completed during the previous reporting period identified the eMagin Z800 3D visor as an optimal interface for the 3D SPACE software. The only drawback to the Z800 visor was that, like most commercial off-the-shelf (COTS) 3D head-mounted displays, it required a specific graphics card in order to produce 3D images, particularly from a laptop computer (the NVIDIA Quadro). The Dell M-90 was one of the few laptops that came with the NVIDIA Quadro and supported generation of 3D images on the Z800. A Dell M90 was purchased and configured for testing of the prototype 2D and 3D software. This configuration was tested in ground-based and parabolic flight experiments, and was shown to be effective. During the project PDR, the IBM 31A was identified as the flight laptop for ISS experimentation. At this point, eMagin Corporation had recently released a firmware upgrade, new visors with embedded firmware upgrades, and a software developer's kit (SDK) for development of custom 3D generation software which could be run on desktops and laptops supportive of 3D graphics, not limited to those with NVIDIA graphics cards. A firmware upgrade was purchased and installed for the previously purchased 3D visor and a 3D visor with an embedded firmware upgrade was purchased for development and testing of the final system configuration. Software Development: Software development efforts for the current reporting period included design and implementation of the 3D graphics generation code, graphical user interface (GUI), individual 2D and 3D task components, and database. The 3D graphics generation required additional programming using a software developer's kit (SDK) from the 3D visor manufacturer, eMagin Corporation, in order to allow 3D graphics to be displayed from a laptop computer without a specified graphics card (i.e., NVIDIA Quadro FX). The GUI was modified to support ease of use with minimal instruction. The initial 2D and 3D task components were developed for parabolic flight testing during the current reporting period and modified based on feedback and observations during testing. During initial testing, data was collected and saved locally to a text file for post hoc analysis. For the flight software, a database was created to save and pre-process data in near real-time. The final flight software is currently being transitioned to ESA. Ground-Based and Parabolic Flight Testing: A parabolic flight experiment conducted on the 43rd ESA Parabolic Flight Campaign (March 26-30, 2006 examined the strength of classical geometric illusions (i.e., reversed T, Muller-Lyer, Ponzo, Pogendorff, Necker's cube) in microgravity. A percent estimate of size or tilt differential was calculated for each figure and compared between normal gravity and microgravity. Results showed that the strength of geometrical illusions based on perspective decreased by 28 to 51% for all subjects in microgravity compared to normal gravity. The depth of a Necker's cube also appeared 28% smaller in microgravity. However these decreases were not found to be significant ($p<0.08$). These results were presented and published in the proceedings of the 57th International Astronautical Congress in Valencia, Spain (Oct 2-6, 2006). An additional parabolic flight experiment was conducted on the 44th ESA Parabolic Flight Campaign (Oct 9-13, 2006) to examine the effects of short-term microgravity on perception of 3 dimensional (3D) objects and distances in 3D scenes, the results of which indicated that, on average, the subjects' perceived height of a 3D cube decreased by about 8% , whereas both its perceived width and depth increased by about 5% in microgravity compared to normal gravity. The relative distance of a 3D cylinder was affected in the same way. All four absolute distance estimations in the sagittal plane were larger (by 29.1 % in average, $p<0.05$) in microgravity compared to normal gravity. The estimations in the frontal plane were unchanged. Preliminary Design Review (PDR) On October 14, 2006, the Preliminary Design Review (PDR) for this experiment was conducted at CNES in Toulouse, France with members of ESA present and the NASA program manager participating via teleconference. During the PDR, and overview and review of the project was presented, a demonstration of the current system was presented, software and hardware requirements for flight equipment were finalized, and the current Experiment Requirements Document (ERD) (ILSRA-2001-088) was reviewed. Experimental Design: Following the PDR, the research team updated Experiment Requirements Document(ERD) (ILSRA-2001-088) and

	submitted the finalized document to ESA. Administrative Tasks: Administrative tasks conducted during the current reporting period included weekly team meetings and teleconferences with the ESA PI, progress documentation and reporting, preparation for the PDR, and travel to and from meetings and experiments.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Papers from Meeting Proceedings	Clement G, Lathan C, Bukley A. "Mental Representation of Spatial Cues in Microgravity." 57th International Astronautical Congress, Valencia, Spain, Oct 2-6, 2006. IAC-06-A1.3.03, 2006. , Oct-2006